Motivate Future Doctors: Evidence from a Large-Scale Survey Experiment in China*

Chunqing Li[1], Xuanxuan Ma[2,3], Hongbin Wu[2], Xiaoyang Ye[4], and Hang Yu[5,6]

[1]Education Management Information Center, Ministry of Education, Beijing 100191, China
[2]National Center for Health Professions Education Development/Institute of Medical Education, Peking University, Beijing 100191, China
[3]School of Public Health, Peking University, Beijing 100191, China
[4]Amazon Group
[5]China Center for Economic Research, Peking University, Beijing 100871, China
[6]National School of Development, Peking University, Beijing 100871, China

Email: cql@pku.edu.cn (Li), wuhongbin@pku.edu.cn (Wu), xiaoyang.ye26@gmail.com (Ye); hyu@nsd.pku.edu.cn (Yu)

*Correspondence:
Hongbin Wu
National Center for Health Professions Education Development/Institute of Medical Education, Peking University, No. 38 Xue yuan Rd, Haidian District, Beijing 100191, China
Phone number: +86 15120099396
Email: wuhongbin@pku.edu.cn

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** This study has been approved by Peking University Institutional Review Board IRB00001052-20069 and registered at AEA RCT Registry (AEARCTR-0006977). Equal authorship applies. Correspondence: (Wu) wuhongbin@pku.edu.cn, Tel: 15120099396 ; Email address: cql@pku.edu.cn (Li), doublexuan@pku.edu.cn (Ma), xiaoyang.ye26@gmail.com (Ye),hyu@nsd.pku.edu.cn(Yu). This work was supported by National Natural Science Foundation of China, and the grant number is 72174013. The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript. We express our gratitude to the National Center for Health Professions Education Development for the support. The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Abstract

This paper examines the impact of motivation on medical students' career commitment. We designed and analyzed the China Medical Student Survey, covering nearly half of the country's medical student population, and identified significant attrition risks. We embedded a randomized experiment in the survey, testing five motivational behavioral nudges. Results show that interventions appealing to intrinsic motivation, such as highlighting the significance of the profession, emphasizing respect for doctors, and using altruistic role models, significantly boosted career commitment, particularly among students with initially low intrinsic motivation. By contrast, extrinsic motivators, including information on salary prospects and peer retention rates, had smaller effects.

Keywords: Intrinsic and extrinsic motivation, career choice, medical student, experiment, behavioral nudge

JEL Classification J24 D91 I23 O15

1. Introduction

The past few decades have seen a steady rise in human resources for health (HRH) globally, however, we still fall short of all types of HRH to achieve Universal Health Coverage, especially in the developing regions (Haakenstad et al., 2022). HRH in China has been among the fastest-growing in the last few decades. For example, the density of medical doctors, the core component of HRH, increased from 12.7 to 25.2 per 10,000 population from 2000 to 2021.[1] The annual number of medical graduates increased more than three-fold during the same period.[2] However, this substantial expansion is not without concerns. Teaching resources are stretched thin (Hou et al., 2014; Wang, 2021), and we saw an ever-higher attrition of medical graduates. According to a survey of new medical graduates by the National Center for Health Professions Education Development (NCHPED) in 2019, fourteen percent of the bachelor's degree earners from medical school (after five years of undergraduate medical education) did not choose to continue their medical careers (Ye et al., 2022).

At one level, the high dropout rate tempers the country's goal of bridging its supply gap, leading to substantial efficiency loss in the economy.[3] At a deeper level, the high attrition rate reflects insufficient career commitment among medical graduates overall. Beyond the mere drop-out numbers, some medical graduates unpassionately choose to become doctors. This is concerning, as the medical profession features asymmetric information and requires a prosocial sense of career identity (Arrow, 1963). Doctors with less commitment to their profession will eventually lead to more deadweight loss in the healthcare market.

In this study, we surveyed and conducted experiments with medical students in China to examine their career choices. Undergraduate medical education is the first and only entry point for young talents to become doctors in China. Due to the closed system, medical students define the pool for all future doctors. We designed and analyzed the China Medical Student Survey (CMSS), the largest medical student survey in China, covering almost half of all current medical students in undergraduate medical education in the country. We documented that a baseline intrinsic motivation index is highly correlated with a student's intention to pursue a medical career. It has stronger prediction power of a student's career choice than other factors such as satisfaction with college admission results, college experience, or a baseline extrinsic motivation index (Appendix Figure A1).

[1] WHO Data Portal: https://apps.who.int/nhwaportal/. last accessed June 20, 2024.
[2] Wang (2021) estimates that the number of medical graduates with a bachelor's degree or higher increased from around 30,500 in 2000 to more than 94,600 in 2018.
[3] Frenk et al. (2022) estimated the cost per graduate in China to reach 41,000 USD by 2018.

To establish a causal relationship between motivation and career choice and explore scalable methods to improve the career commitment of medication students, we embedded an experiment into the 2021 CMSS. One of five (or no) text-graphic information treatments was randomly applied to survey respondents, each targeting a different dimension of individuals' motivation. The first three treatments focused on intrinsic motivation: Treatment 1 (T1) stressed the significance of the medical profession to society, Treatment 2 (T2) highlighted the social respect for doctors that dates far back in history and lasts to the present day, and Treatment 3 (T3) primed doctors' altruistic identity and introduces a doctor role model for the students. The remaining two treatments were based on extrinsic motivation or social comparison: Treatment 4 (T4) made salient doctors' high and fast-growing salaries and Treatment 5 (T5) presented the retention rate (in becoming doctors) of peer medical students.

We find each of the five treatments significantly increased students' reported commitment to becoming doctors. Interventions targeting intrinsic motivations generate more substantial effects. Treatment arms Significance of the profession (T1), Social respect for doctors (T2), and Altruism identity and role model (T3) increased students' self-reported career commitment by 1.8, 3.0, and 2.8 percentage points, respectively, compared to the control group (76.8 percentage points). The salary information (T4) and the peer retention rate (T5) also had significant positive effects but of a smaller size.

We provide further evidence to show that the change in intrinsic motivation is the behavioral mechanism behind the three largest treatment effects. First, we explored treatment heterogeneity across groups with varying levels of baseline motivations. T1, T2, and T3 treatment effects became monotonically larger as a subsample's baseline intrinsic motivation decreased. There was no such treatment heterogeneity pattern by baseline extrinsic motivation nor treatment heterogeneity by baseline intrinsic motivation for T4 and T5. Second, we collected a wide range of respondents' post-treatment subjective beliefs, such as career aspirations, study plans, confidence, and emotional attachment to and perceived social values of the career. The three treatments targeting intrinsic motivations, but not the other two, consistently improved all aspects above.

This paper contributes to the literature on how motivations shape career choices. Motivation's role in individual behavior has long been regarded in theoretical work in economics (Akerlof and Kranton, 2000; Bénabou and Tirole, 2003; Besley and Ghatak, 2005). Bénabou and Tirole (2006) proposed a model that provides a conceptual structure for later works where an agent's motivations are classified as extrinsic (related to financial incentives), reputational (related to social status), and intrinsic (motivations without reward to monitoring). Since then, economists have applied the framework to understand career choice, especially in occupations with public features, and

developed an extensive empirical literature.

A thread of this literature studies how different careers select individuals of varying motivation types. In their review, Finan et al. (2017) summarized an observation that public sector workers tend to be more intrinsically motivated. A series of experimental studies further investigated the selection effects by varying job (perceived) features. Dal Bó et al. (2013) did not find higher financial incentives crowd out public service motivations among Mexico's public community agents. Ashraf et al. (2020) found some evidence that external incentives crowed out prosocial incentives in recruiting nurses in Zambia, but only among low-talented individuals. Deserranno (2019) documented that financial incentives discouraged prosocially motivated candidates from applying for health worker jobs in Uganda.

We take a different experimental approach. This study uses behavioral nudges to alter individuals' motivations to encourage entering careers essential to public welfare. It fits contexts where the job features are widely known (e.g., doctors), and the pool of candidates is fixed (e.g., medical students). A few recent studies adopted a similar approach. Ajzenman et al. (2023) experimented with teachers in Peru and found that both priming altruism identity and highlighting monetary rewards can encourage them to apply to disadvantaged schools. Gulzar and Khan (2024) found in Pakistan that emphasizing prosocial motives rather than personal returns encouraged individuals to run for political offices.[4]

Our study adds to this line of literature from several angles. First, we extend research to the medical education context. We find that nudges appealing to intrinsic career motivations most effectively encourage medical students to serve as doctors after graduation. Second, we collected detailed information about the subjects' psychology and motivation at the baseline. This allows us to conduct an in-depth analysis of the treatment heterogeneity. Finally, the scale of our study is substantial. We experimented with more than 200,000 medical students in China, who would become a large proportion of the country's future doctors in the next decades. The sample size allows us to measure the treatment effects of light-touch interventions accurately and makes the empirical findings readily policy-relevant.

This study also relates to the literature encompassing how motivations are enacted and affect behavior. Theoretical work primarily originated from the psychology literature. Self-determination theory provides a conceptual framework that maintains innate psychological needs for autonomy, competence, and relatedness as sources for motivation (Deci and Ryan, 1985, 2000; Ryan and Deci, 2000). Grant (2008a, b) stressed the "significance of task significance" and explained how

[4] Some other studies, though not taking career choice as the primary outcome, adopted similar designs to change motivation by behavioral interventions. For example, Fracchia et al. (2023) encouraged community health volunteers' performances by providing honorific rewards and showing videos to increase the perceived significance of their jobs.

establishing connections between the job and those who benefit from the job motivates prosociality and facilitator of behavioral change. Merton (1957) and Morgenroth et al. (2015) analyzed the motivational effect of "role models" as they can act as behavioral models, represent the possible, and be inspirational.

Many behavioral interventions are proposed and empirically tested in labor market settings, such as highlighting the meaning of the job (Cassar and Meier, 2018; Nikolova and Cnossen, 2020), enacting prosociality through relational mechanisms (Grant and Shandell, 2022), priming social respect and reputational reward (Ellingsen and Johannesson, 2007; Kosfeld and Neckermann, 2011; Gallus and Frey, 2015), and introducing role models (Lazowski and Hulleman, 2016; Serra, 2022). The effectiveness of such interventions varies by context, and there is a lack of systematic comparison among the interventions.

We contribute by comparing multiple behavioral interventions that target different motivations in one setting. The five treatment arms in our experiment—Significance of the profession, Social respect for doctors, Altruism identity and role model, Salary level, and Peer retention rate—each target a different aspect of individual motivation for pursuing a doctor's career. We find that the three interventions enacting intrinsic motivations more effectively encourage medical students to become doctors. Financial incentives and nudges that appeal to social comparison were not as effective.

Lastly, this study also relates to a small but rising literature on "brain drain" in healthcare. The attrition of graduates is a new form of "brain drain" from the medical profession that could bring substantial social welfare loss. Previous research has drawn attention to international factors that may affect the healthcare labor force in developing countries through the emigration of healthcare professionals (Bhargava and Docquier, 2008; Abarcar and Theoharides, 2024). We studied an internal factor - the lack of motivation. We provide evidence that motivations can be nurtured at the individual level.

China's medical education system has long been criticized both inside and outside for focusing too much on technology and knowledge and neglecting humanity and ethical education (Hou et al., 2014; Wang, 2021). From a policy perspective, our study shows that nurturing motivation through intervention is feasible and plausible. This implication goes beyond the context of China. Developing countries across the globe, with even more limited medical education resources but more pressing demand to expand their healthcare supply, should stay alert to motivating their young talents.

2. Medical Education in China and the China Medical Student Survey (CMSS)

Medical education in China starts at the undergraduate level. Upon completing high school, aspirants aiming to become doctors enter medical schools based on their National College Entrance Examination (NCEE) scores. Most students are first admitted to a 5-year undergraduate medical program, where they undergo education spanning general studies, basic medical training, clinical education, and clerkship. Students then proceed to graduate study or residency training through an application process (Zhu et al., 2016).[5] All medical students must take the National Medical Licensing Examination (NMLE) one year after graduation from undergraduate medical education. Only those who have passed the NMLE can obtain the physician qualification certificate.

Undergraduate Medical education in China is a closed system, and admission to medical school after high school is almost the only entry point. Furthermore, a bachelor's degree from a medical school is a prerequisite for registering for the NMLE (Wang, 2021). As a result, the number of admissions to medical programs can approximate the upper limit on the number of future qualified doctors (Hou et al., 2014).

The China Medical Student Survey (CMSS) is an annual national survey conducted by the National Center for Health Professions Education Development (NCHPED). It was first piloted in 2019. NCHPED is the institute established by the Ministry of Education and the National Health Commission, with the responsibility of promoting the reform and development of medical education in China. One of its key functions is to issue the standards for undergraduate medical education and to implement the accreditation of medical education China and is thus well-regarded by medical schools. NCHPED announced the survey among medical schools. Each participating school designated a program manager who has undergone specific training regarding the survey process. The program managers at these schools invited all medical students at their schools to participate in the survey. The survey is then distributed to students online by official venues, while students' participation is voluntary. Participants self-administer the survey on a specially developed online platform. The annual CMSS survey collects self-reported data on student demographic data, pre-college backgrounds, curriculum satisfaction, teaching service feedback, learning encounters, and future career plans.

[5] There are two special programs where students automatically proceed to the next stage without going through application processes. The "8-year" track is tailored for prospective physician-scientists, where the graduates will be granted bachelor's and doctorate degrees after eight years of training (Wu et al., 2021) the "5+3" track combines the 5-year medical undergraduate education and 3-year medical master (MM) program.

2.1. Study Sample from CMSS

This study utilizes the 2021 sample of CMSS. In 2021, 63 percent, or 123 out of 193, medical institutions in China offering undergraduate programs partake in this survey. This study focused on three major programs in China, including the rural 5-year program,[6] 5-year program, and 8-year program, compiling data from 213,326 students across the 123 participating medical institutions. This sample covers about 50% of all undergraduate-level medical students in China.

The sample students are, on average, 21 years old and are equally distributed across the five undergraduate years of medical study. Fifty-six (56) percent are female, and thirty-nine (39) percent are the only child at home. The students come from better family backgrounds than the average age cohort in China. The median student in our sample comes from households with annual income between 30-80 thousand yuan (4,200-11,000 USD by PPP). About a quarter of their father and mothers have a college degree. In particular, thirty-two (32) percent of the sample had at least one parent working in medicine or other health-related industries. The vast majority (or 99 percent) entered their current college through the National College Entrance Examination. The proportion of students from schools in the country's eastern, central, and western regions is 38.4%, 28.0%, and 33.7%, respectively. Detailed summary statistics can be found in Appendix Table A1.

2.2. Baseline career commitment and correlated factors

In collaboration with NCHPED, we included a rich set of questions in the 2021 CMSS to collect data on students' per-college and college experiences, opinions, attitudes, psychological features, and baseline career commitment (before any interventions were applied).

We summarized key student characteristics into four indices: their intrinsic and extrinsic motivations for choosing the medical major, attitudes toward their college admission results, and attitudes toward their college experiences. Intrinsic motivation index measures the degree to which a student chooses a medicine major due to prosocial preferences, intellectual interests in medical science, or admiration of medical professions. The extrinsic motivation index measures driving factors such as the lucrative income prospect of a medical career or the social resources it brings. The major satisfaction index measures how satisfactory a student was to be admitted to the current program and whether they were familiar with the discipline before entering college. The college experience index measures students' subjective evaluation of their studies, personal lives, and interactions with professors or schoolmates. The construction of each index is described in depth

[6] Starting in 2010, to address the shortage of medical and health professionals in rural and remote areas of central and western China, the Chinese government introduced the policy of "The rural-oriented tuition-waived medical education program" in 5-year program, also known as the "rural 5-year program".

in Appendix B.

We measured baseline career aspiration through the following question:

Question-Pre: "After completing your undergraduate studies, do you wish to become a doctor?"

Eighty-three (83) percent of the respondents answered "Yes," while 14 answered "Unsure," and 3 answered "No." Table 1 regresses students' answers to this question against four measures of baseline characteristics described above.

We show that a student's career aspiration strongly correlates with all four indices, most substantially with intrinsic motivation. In regression analyses that control for individual characteristics and school fixed effects, one standard deviation increase in the intrinsic motivation index is associated with a 9.6 percentage point increase in reported wanting to become a doctor (answered "yes" to the baseline career aspiration question above) (Table 1). Such a strong correlation set the stage for us to explore further whether intrinsic motivation can causally affect students' career choices.

3. Experiment Design

As the second-last section of the survey, we incorporated an experimental module into the 2021 CMSS. The online survey platform's algorithm automatically and randomly assigned each individual respondent to one of five treatment groups or a control group. Appendix Table A1 reports the balance test.

After answering questions about the individual characteristics described above, survey respondents were randomly presented with one of the following five text-graphic messages (treatment groups) or no such messages (control group). The exact intervention message can be found in Appendix C.

T1: Significance of the profession. The intervention is an adaptation of a speech delivered by Dr. Qimin Zhan, an esteemed leader in the medical science community. In it, he called the healthcare professionals pivotal national resources, cited the fight against COVID-19 to emphasize the key role of health professionals in society's well-being, and called for a culture that honors and supports healthcare providers.

T2: Social respect for doctors. This intervention presents two photos that depict a child patient and a doctor bowing to each other, one set in the Qing Dynasty in the 19th century and the other in contemporary times. The accompanying text stresses the respect and care between doctors and patients.

T3: Altruism identity and role model. This intervention presents an appraisal paragraph of

the heroic performance of healthcare workers in combating COVID-19 viruses. It features Dr. Zhong Nanshan, the lead medical scientist during both the SARS outbreak in 2003 and the COVID-19 outbreak in 2020 in China, for his humble gestures and dedicated efforts to the profession.

T4: Salary level. This intervention presents a statistical graph based on the China Hospital Salary Survey, depicting doctors' average salaries in 2019 by city tiers in China. The graph highlights the national average of 185,000 yuan (44,000 USD by PPP), which is 4.35 times the average salaried job in China and is growing at 4% annually.

T5: Peer retention rate. This intervention presents a statistical graph showing medical undergraduate students' intention to become doctors after graduation by gender and program type —the national average features eighty-seven (87) percent.

The first three treatments aim to appeal to different aspects of students' intrinsic motivation. T1, by stressing the meaning of the medical profession to society, draws from the theory that a job's significance affects individual motivation (Deci and Ryan, 1985; Ryan and Deci, 2000; Cassar and Meier, 2018). T2 establishes a concrete and humanized connection between doctors and benefiting patients. The theory predicts that such a connection enhances prosocial motivations (Grant and Shandell, 2022). Both T1 and T2 also stress the social respect associated with the medical profession, which facilitates a sense of competence - an important source of intrinsic motivation. T3 appeals to the impact of role models (Morgenroth et al., 2015; Serra, 2022) that medical students can relate to and be inspired by. T4 and T5, on the other hand, present factual information as a comparison.

T4 can potentially invoke extrinsic motivation for becoming a doctor as it highlights the high economic return of this profession.[7] In theory, the direction of T4's treatment effect should depend on the participants' prior belief of the salary level. However, we cannot directly test this hypothesis as pilot studies suggest that the population of interest will not report reliable measures of income expectation for the future. We expect T4 will, on average, increase students' willingness to become doctors since it stresses the high salary level compared to other professions.[8]

T5 aims to help participants understand the career intentions of their peers. It can help students contextualize their career goals and make informed decisions regarding their future paths in the medical profession. T5 is predicted to reduce dropout intention among those who overestimate the dropout rate to two potential channels. First, it may invoke individuals' tendency to conform to the

[7] There is a rich literature on how extrinsic incentives may affect intrinsic motivation or altruistic behavior (Bénabou and Tirole, 2006). A crowding-out effect is found in some contexts (Frey and Oberholzer-Gee, 1997; Mellström and Johannesson, 2008; Deserranno, 2019), but in others (Lacetera et al., 2012; Ashraf et al., 2020; Goette and Stutzer, 2020).
[8] The average salary level depicted in T4, 185,000 yuan, was much higher than the average income of the households the students were from, between 30,000 to 80,000 yuan.

norms (Benabou and Tirole, 2011). Second, a higher-than-expected retention rate provides new information to students regarding the desirability of the medical profession. We will explore the treatment heterogeneity based on the prior belief of the peer retention rate.

Following randomization, each treatment group had between 16.6% and 16.9% of the sample, with 16.8% assigned to the control group. From the background technical data, the average time spent on the information page is 12 seconds – it is long enough for individuals to process the message while short compared to the overall length of the survey (which is 23 minutes). The treatment and control groups do not significantly differ in overall survey time. Respondents from different treatment groups spend no significant differences in time on the information page.

After the messages were shown (or lack thereof in the control group), the survey proceeded to its last section, where participants were asked about their post-graduation plans and attitudes about their professional identity. Among them, we assessed participants' aspirations to become doctors in the last section as our primary outcome of interest. Specifically, the following question:

Question-post: "Please evaluate if the following comment matches you: I want to become a doctor after graduation. Choose one answer from Strongly agree, agree, neutral, disagree, strongly disagree."

In our primary specifications, we construct a binary variable that takes values 1 for "Strongly agree" and "Agree" as "commitment to a career in medicine" and 0 otherwise. Appendix Table A2 replicates the primary analyses while assigning "neutral" to the other side.[9]

Following the Pre-Analysis Plan (Wu et al., 2022), we use the ordinary least squares (OLS) regression to estimate the effects of information interventions on medical students' dropout intention. The principal regression model is presented below:

$$Y_{i,s} = \beta_0 + \sum_{i=1}^{5} \beta_i Treat_i + X_{i,s}\gamma + \mu_s + \varepsilon_{i,s} \qquad (1)$$

Here, $Y_{i,s}$ is a binary variable indicating career commitment (described above) for individual i at school s. $Treat_i$, with i ranging from 1 to 5, indicates the assignment of individual i to specific information intervention groups. $\mathbf{X}_{i,s}$ encompasses control variables, including individual attributes such as gender, ethnicity, political affiliation, family details, and educational metrics. In particular, we controlled for the baseline career aspiration measured by Question-pre before any

[9] The format of the question slightly deviates from what was originally in the Pre-analysis Plan, where students were only allowed to give a binary "yes" or "no" answer. We made the modification because early pilots find that restricting answers to yes or no leads nonnegligible share of missing values. Feedback from pilot subjects also suggested potential misreport problems due to varying subjective interpretations of the question. The final implemented question, Question-Post, allows more flexible answers. We coded the answers to a binary variable to fit into the pre-specified regression equation. In Appendix Table A2, we show that our results are robust to different methods in recoding a 5-scale Likert answer to a binary variable.

intervention was applied. μ_s signifies school fixed effects. The coefficients β_1 through β_5 determine the average treatment effects.

4. Results of the Experiment

4.1. Main Results

Table 2 reports the main regression results.

Column (1) shows the main results without any controls and column (2) reports results from the primary regression model with full controls. The coefficients are stable across specifications. All treatment effect estimates are positive and significant, suggesting information treatment that appeals to motivations indeed increases career commitment. The coefficient on the three intrinsic-motivation-targeted interventions is of the largest magnitude, each causing a two to three percentage points increase in self-reported commitment to a career in medicine. The two other interventions, "Salary level" and "Peer retention rate," had smaller average effects, each around one percentage point.

Column (3) explores potential treatment heterogeneity for T4. As we do not have a reliable prior belief of doctors' salary level from participants, we explore treatment heterogeneity by household income, hypothesizing that students from less economically affluent families will be more motivated by the high salary level of doctors. However, we did not find evidence of treatment heterogeneity. Nor did we find significant treatment heterogeneity of T5 by prior belief peers' career choice (Column 4).

The binary primary outcome variable is constructed from a 5-scale Likert question on the willingness to become a doctor.[10] Figure 1 explores the original answer distribution in more detail to help us understand on which margin each treatment impacted. In Figure 1, we plot the effects of each treatment on the probability of giving each of the five answers to Question-post. Across the board, every treatment reduces the probability of "not committed" answers (strongly disagree, disagree, or neutral to become a doctor), and increases the probability of high commitment (strongly agree to become a doctor).

If we focus on the treatment effect measured by absolute share change (percentage points), all three interventions that target intrinsic motivations are most effective in the middle segment of the distribution. They pushed the undetermined or moderately committed students (neutral or agree to become a doctor) to become highly committed (strongly agree to become a doctor). The two

[10] Given that the 5-scale Likert measure is beyond the scope of the Pre-Analysis plan, the analysis of treatment effects across the distribution should be considered exploratory and interpreted with caution.

extrinsic motivation information interventions have less treatment effect overall. Their effects are more spread out across the full distribution of career commitment.

4.2. Treatment heterogeneity by baseline motivations

If T1 through T3 works by stimulating intrinsic motivation, they should be more effective on students with lower baseline intrinsic motivation. This is because the lack of intrinsic motivation is more likely to be their binding constraint to choose a career in medicine. This prompts an exploration of the heterogeneous effects of our intervention based on motivation types and distribution.

To explore treatment heterogeneity, we rerun Equation (1) in each subsample defined by the baseline intrinsic and extrinsic motivation index quintiles. Figure 2 plots the treatment effect estimates of each intervention by baseline intrinsic motivation index quintile subsamples. The three intrinsic-motivation-targeted interventions exhibit strong treatment heterogeneity – these interventions are monotonically more effective in reducing dropout intention among students who are less intrinsically motivated at the baseline. For instance, the "Social respect for doctors" (T2) intervention increases career commitment by 5.3 percentage points in the quintile with the lowest baseline intrinsic motivation but only 0.5 percentage points in the highest quintile.

We formally tested the treatment heterogeneity pattern in panel B. It reports the p-values of the equality tests of each treatment arm's effect between each pair of subsamples (defined by baseline intrinsic motivation index quintiles). As we can see, T1's effects significantly differ (at a 5 percent level) between subsamples that are at least two quintiles apart (Q1 vs. Q4 or Q2 vs Q5), sometimes one quintile apart (Q2 vs Q4, Q3 vs. Q5). T2's effect significantly differs between subsamples at least 1 quintile apart, sometimes even between neighboring quintiles. T3's effect significantly differs between subsamples at least two quintiles apart. On the contrary, none of T4 or T5's treatment effects exhibit significant heterogeneity between subsample pairs defined by baseline intrinsic motivation.

As a placebo test, we replicated Figure 2 but redefined sub-samples by baseline extrinsic motivation index quintiles in Appendix Figure A2. There are no consistent treatment heterogeneity patterns like in Figure 2.

4.3. More evidence on mechanisms

Beyond the pre-specified primary outcome of interest, we collected students' post-intervention attitudes on a wide range of statements, all evaluated on a 5-point Likert scale. Some of the

statements are alternative framings of the primary outcome that can assess the robustness of the findings. Others provide insight into students' perceptions of the medical profession, such as its social significance, students' confidence and aspiration of the profession, and students' study plans for taking action.

Table 3 showcases the OLS regression results of these dimensions against five intervention indicators following Equation (1). Notably, three interventions targeting intrinsic motivations – Significance of the profession, Social respect for doctors, Altruism identity and role model – strongly impacted all dimensions. In contrast, the interventions focusing on salary level and peer retention rate revealed mixed results, with some dimensions not reaching statistical significance.

Consistent with the main results (Table 2 Column 2), T1 through T3 have positive impacts ranging from two to four percentage points on two alternative statements regarding career commitment, Columns (1) and (2). At the same time, the effects of T4 and T5 are substantially smaller and sometimes insignificant. When a student is more strongly motivated to pursue a career in medicine in the future, we expect her to take action in promoting their study in college. Columns (3) and (4) confirmed this prediction. T1 through T3 cause students three to four percentage points more likely to (report) plan to participate actively in college study events. T4 and T5 also have some positive effects but are significantly lower in magnitude.

Columns (5) to (11) explore how our information treatments affect multiple dimensions of students' attitudes toward the medical profession. Attitude changes are a potential mechanism that eventually affects career decisions. We find that T1 through T3 significantly improved students' aspirations for a career as a doctor and for the healthcare industry (columns 5 and 6), making students more appreciative and taking more pride in the medical profession (columns 7 and 8). They boost the understanding of the meaning of a doctor's job (columns 9 - 10), and recognition of the social value of the profession (column 11). T4 and T5's treatment effects on these aspects are smaller or insignificant, a pattern that trails the main results well.

5. Conclusion

In this study, we conducted a large-scale survey experiment among medical students in China. Light-touch informational interventions that appeal to intrinsic motivations can nudge medical students to be more committed to the doctor's career. Text-graphic messages highlighting the significance of the medical profession, appraising the social respect dedicated to the profession, and introducing altruistic role models of the profession are the most effective. Further analyses on treatment heterogeneity and other secondary outcomes of interest confirmed that the information interventions worked through enacting intrinsic motivations of various dimensions.

This study helps us understand the causal relationship between motivation and career choice. It also provides evidence that motivations can be nurtured and, if nurtured correctly, can contribute to solving real-world problems. China has yet to close its gap in building sufficient human resources in healthcare, especially in its economic and demographic transition era. Training and motivating a committed workforce in medicine is critical to the population's health and the stability of its economy. Our study showed the insufficient career commitment among medical students in China and proposed potential policy directions. More humanity, social science, and ethical education are called for in training health workers. Nudges like ours in the study are low-cost and can be valuable supplements to the education of medical professionals.

The lack of human resources in healthcare is an even more severe challenge in lower-income parts of the world. This study suggests that countries seeking to expand their health workforce supply rapidly should pay attention to the motivational aspect of medical education. The evidence of the effectiveness of targeted informational interventions provides valuable insights for educators, policymakers, and medical institutions. As we navigate the complexities of medical education, understanding and addressing motivational issues are crucial steps toward building a resilient and dedicated healthcare workforce. We also provide examples of effective nudges to enhance commitment and mitigate attrition in the medical field, inspiring future doctors and contributing to the broader goals of healthcare systems in providing high-quality care.

Conflict of interest statement

We declare no competing interests.

Data Availability Statement

The parameters that we used in our model (text, tables, figures, models, and appendices) are available on reasonable request from the corresponding author under certain conditions (with the consent of all participating centers and with a signed data access agreement).

Figures and Tables

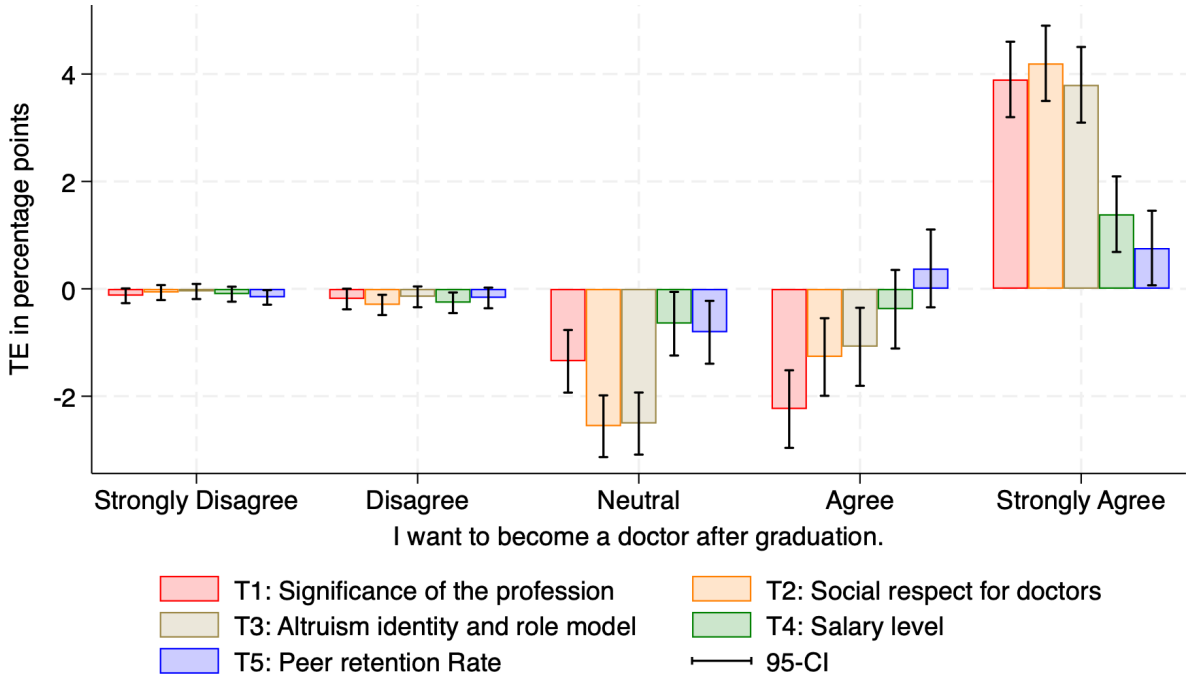
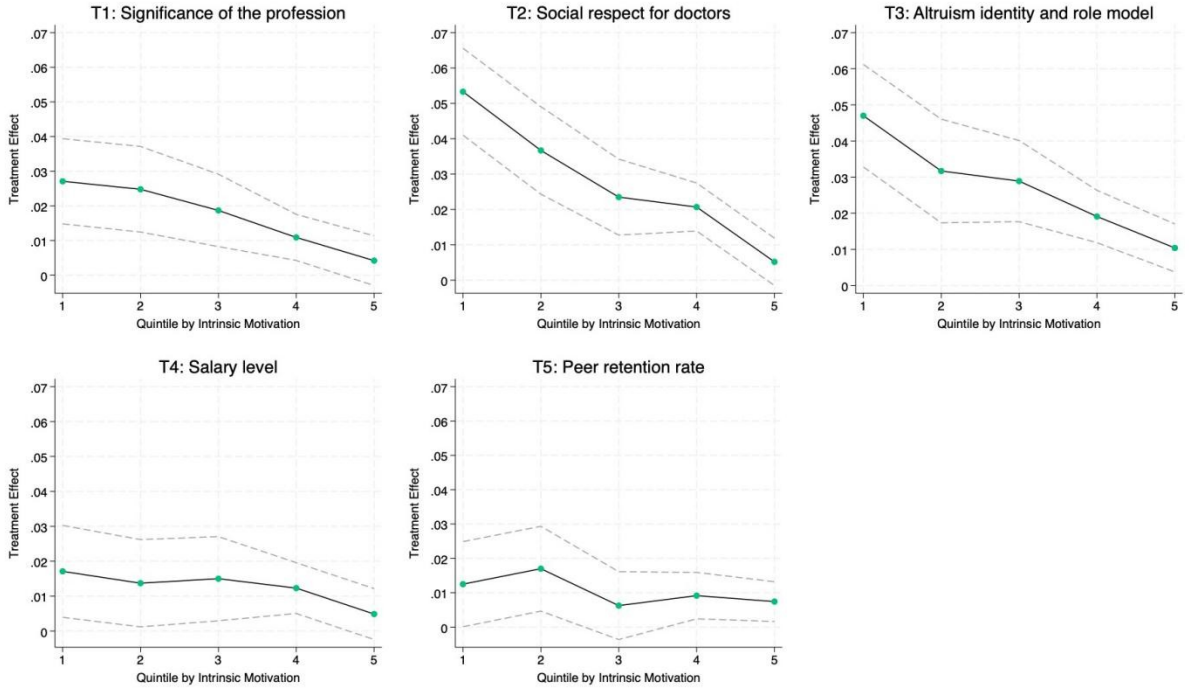


Fig. 1. Treatment effect on career commitment distributions

Panel A: treatment effects by intrinsic motivation quintiles



Panel B: p-values of equality test between quintile pairs

T1: Significance of the profession

	Q1	Q2	Q3	Q4
Q2	.801			
Q3	.346	.493		
Q4	.027**	.046**	.182	
Q5	.002***	.004***	.022**	.168

T2: Social respect for doctors

	Q1	Q2	Q3	Q4
Q2	.049**			
Q3	.001***	.127		
Q4	.000***	.024**	.671	
Q5	.000***	.000***	.002***	.002***

T3: Altruism identity and role model

	Q1	Q2	Q3	Q4
Q2	.142			
Q3	.082*	.773		
Q4	.000***	.123	.183	
Q5	.000***	.010***	.002***	.076*

T4: Salary level

	Q1	Q2	Q3	Q4
Q2	.71			
Q3	.809	.893		
Q4	.535	.861	.722	
Q5	.103	.259	.148	.108

T5: Peer retention rate

	Q1	Q2	Q3	Q4
Q2	.656			
Q3	.432	.211		
Q4	.611	.265	.617	
Q5	.460	.148	.842	.684

Fig. 2. Treatment heterogeneity by baseline intrinsic motivation index

Table 1 Predictors of baseline career commitment

	Baseline Career Commitment				
	(1)	(2)	(3)	(4)	(5)
Intrinsic motivation	0.096***				0.078***
	(0.002)				(0.001)
Extrinsic motivation		0.034***			-0.007***
		(0.001)			(0.001)
Major satisfaction			0.081***		0.043***
			(0.002)		(0.002)
College experience				0.057***	0.019***
				(0.002)	(0.001)
Other Controls	YES	YES	YES	YES	YES
School FE	YES	YES	YES	YES	YES
R^2	0.159	0.110	0.135	0.123	0.170
Observations	213,326	213,326	213,326	213,326	213,326

Notes: Mean baseline career commitment is 83.2%. The four explanatory variables are standardized within the sample. The coefficients should be interpreted as the change in outcome associated with one standard deviation change in the explanatory variable. Standard errors are clustered at the school level, shown in the parentheses. Other controls are listed in Table A1. * significant at 10%, **5%, and ***1%.

Table 2 Main results

	Endline Career Commitment			
	(1)	(2)	(3)	(4)
T1: Significance of the profession	0.017***	0.018***	0.018***	0.017***
	(0.003)	(0.003)	(0.003)	(0.003)
	[0.001]	[0.001]		
T2: Social respect for doctors	0.029***	0.030***	0.030***	0.030***
	(0.004)	(0.003)	(0.003)	(0.003)
	[0.001]	[0.001]		
T3: Altruism identity and role model	0.027***	0.028***	0.028***	0.028***
	(0.003)	(0.003)	(0.003)	(0.003)
	[0.001]	[0.001]		
T4: Salary level	0.010***	0.012***	0.014***	0.012***
	(0.004)	(0.003)	(0.004)	(0.003)
	[0.008]	[0.001]		
T5: Peer retention rate	0.011***	0.011***	0.011***	0.010***
	(0.003)	(0.003)	(0.003)	(0.003)
	[0.001]	[0.001]		
Low family income			-0.011***	
			(0.002)	
Low family income			-0.002	
× Salary level (T4)			(0.004)	
Overestimate dropout				-0.037***
				(0.003)
Overestimate dropout				0.004
× Peer retention rate (T5)				(0.005)
Baseline Career Commitment	NO	0.555***	0.555***	0.551***
	NO	(0.005)	(0.005)	(0.005)
Control group mean	0.768	0.768	0.768	0.768
Other Controls	NO	YES	YES	YES
School FE	NO	YES	YES	YES
R^2	0.001	0.306	0.306	0.307
Observations	213,326	213,326	213,326	213,326

Notes: Standard errors in parentheses are clustered at the school level. The MHT-adjusted p- values are reported in squared brackets. Multiple hypotheses testing is conducted following List et al. (2019). Following Pre-Analysis Plan, control variables include indicators for gender, ethnicity, political affiliation, an only child, grade, enrolled program, having doctors in the family, senior high school type, ideal career, higher parents' expectation, career training participation, and father's ISEI (International Socio-Economic Index) level, mother's ISEI level, father's education level, mother's education level, family income level, NCEE (National College Entrance Examination) score, higher college admission expectation. Significant at *10%, **5%, and ***1%.

Table 3 Secondary outcomes of interests

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Dependent Variable	Medical education has made me more determined to become a doctor.	Being a doctor is my ideal career choice.	I will actively participate in academic and practical activities related to my field.	I will keep up with and follow the latest develop-ments in my professional field.	I am confident that I can become an excellent doctor.	I am optimistic about the future prospects of the healthcare industry.	The profession of a doctor is one of the best professions in today's society.	I feel proud to be a medical student when communi-cating with students from other majors.	I am aware of the responsi-bilities that a doctor should undertake.	Being a doctor is more than just a means of making a living.	Being a doctor is highly respected.
T1	0.024***	0.025***	0.028***	0.033***	0.023***	0.027***	0.032***	0.031***	0.011***	0.014***	0.026***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)
T2	0.038***	0.032***	0.040***	0.046***	0.031***	0.025***	0.034***	0.043***	0.033***	0.029***	0.028***
	(0.004)	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
T3	0.038***	0.030***	0.035***	0.041***	0.028***	0.027***	0.033***	0.041***	0.030***	0.022***	0.030***
	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)
T4	0.007***	0.013***	0.013***	0.025***	0.012***	0.018***	-0.003	0.014***	-0.002	0.019***	0.017***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
T5	0.002	0.009***	0.013***	0.021***	0.008**	0.006*	-0.005	0.014***	0.005*	0.004	0.010***
	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.003)	(0.004)	(0.003)
Control mean	0.723	0.679	0.742	0.685	0.698	0.537	0.500	0.697	0.838	0.417	0.606
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
School FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
R^2	0.186	0.250	0.112	0.123	0.172	0.089	0.092	0.128	0.065	0.054	0.081
Obs.	213,326	213,326	213,326	213,326	213,326	213,326	213,326	213,326	213,326	213,326	213,326

Notes: Control variables are the same as that in Table 2. Standard errors are clustered at the school level shown in the parentheses. Significant at *10%, **5%, and ***1%.

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Motivate Future Doctors: Evidence from a Large-Scale Survey Experiment in China

Appendix

A. Additional Exhibitions

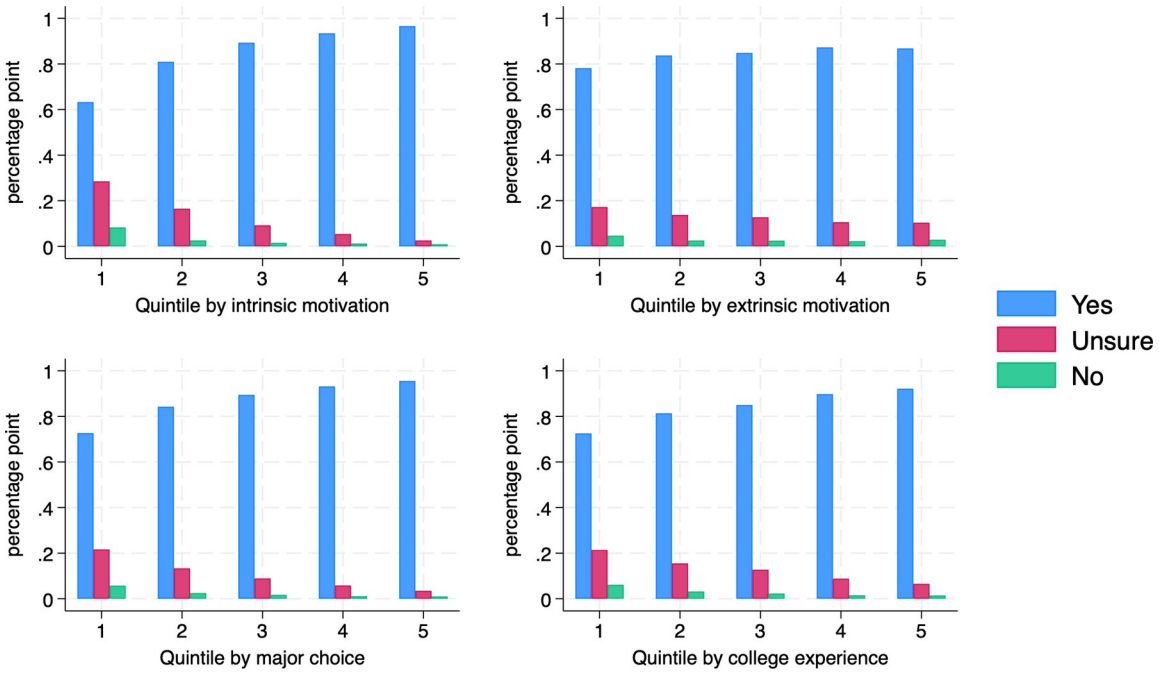
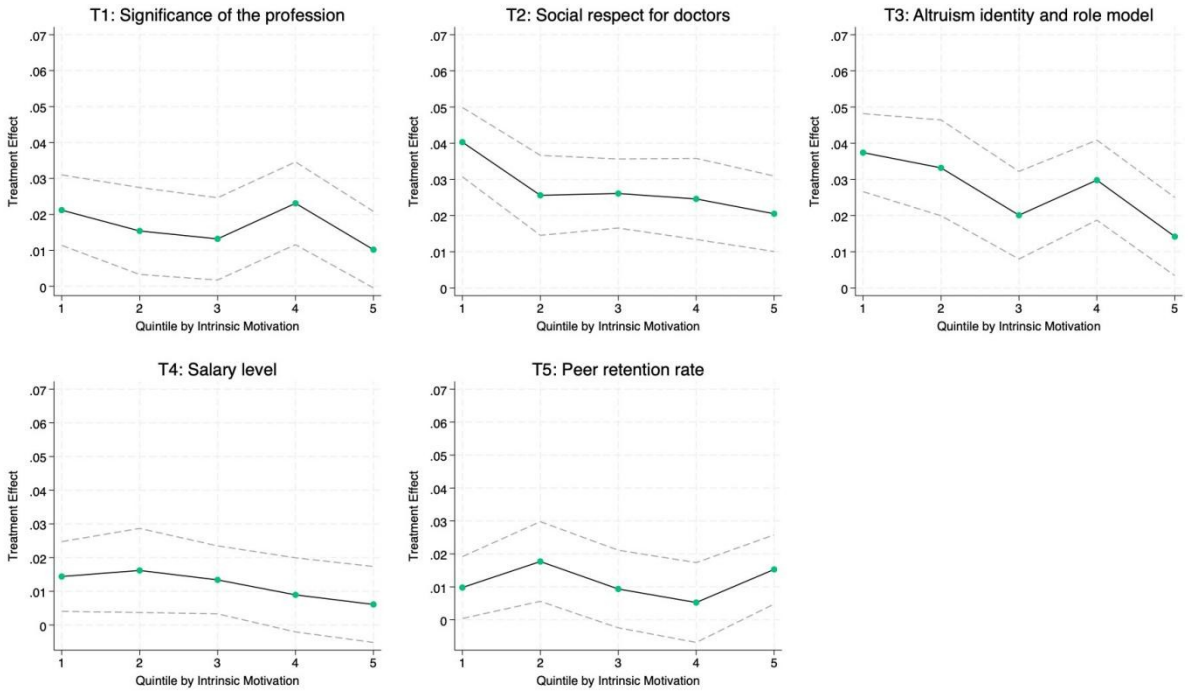


Fig. A1 Baseline career commitment by individual characteristic quintiles

Panel A: treatment effects by extrinsic motivation quintiles



Panel B: p-values of equality test between quintile pairs

T1: Significance of the profession

	Q1	Q2	Q3	Q4
Q2	.513			
Q3	.27	.807		
Q4	.804	.382	.237	
Q5	.168	.5	.716	.068*

T2: Social respect for doctors

	Q1	Q2	Q3	Q4
Q2	.046**			
Q3	.034**	.947		
Q4	.025**	.896	.832	
Q5	.003***	.476	.395	.556

T3: Altruism Identity and role model

	Q1	Q2	Q3	Q4
Q2	.649			
Q3	.019**	.171		
Q4	.333	.713	.25	
Q5	.003***	.02	.509	.027

T4: Salary level

	Q1	Q2	Q3	Q4
Q2	.818			
Q3	.884	.743		
Q4	.449	.397	.58	
Q5	.274	.197	.367	.72

T5: Peer retention rate

	Q1	Q2	Q3	Q4
Q2	.298			
Q3	.954	.359		
Q4	.572	.167	.632	
Q5	.449	.756	.482	.186

Fig. A2 Treatment heterogeneity by baseline extrinsic motivation index

Table A1 Summary statistics and balance

Variable	All	C	T1-C	T2-C	T3-C	T4-C	T5-C
Baseline: doctor	0.832	0.833	-0.003	-0.003	-0.004	-0.005	0.001
	[0.374]	[0.373]	(0.006)	(0.006)	(0.005)	(0.006)	(0.005)
Baseline: unsure	0.136	0.136	0.006	0.000	0.004	0.002	-0.001
	[0.343]	[0.342]	(0.006)	(0.007)	(0.006)	(0.006)	(0.006)
Baseline: dropout	0.032	0.031	-0.012	0.013	0.006	0.015	0.000
	[0.176]	[0.175]	(0.010)	(0.010)	(0.011)	(0.011)	(0.009)
Gender	0.443	0.441	-0.000	0.001	0.001	0.000	0.007*
(Male=1)	[0.497]	[0.497]	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Ethnicity	0.885	0.886	-0.005	-0.005	-0.003	-0.001	-0.003
(Han ethnic=1)	[0.319]	[0.318]	(0.005)	(0.006)	(0.005)	(0.006)	(0.006)
CPC members	0.098	0.098	-0.002	-0.006	0.005	0.004	0.002
	[0.297]	[0.297]	(0.007)	(0.006)	(0.006)	(0.007)	(0.006)
Only child	0.393	0.397	-0.006**	-0.000	-0.002	-0.009**	-0.006*
	[0.488]	[0.489]	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)
First-year	0.247	0.246	-0.002	0.007	0.003	0.001	0.005
	[0.431]	[0.430]	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
Second-year	0.202	0.204	0.003	-0.001	-0.006	-0.009*	-0.003
	[0.402]	[0.403]	(0.004)	(0.005)	(0.004)	(0.005)	(0.005)
Third-year	0.223	0.222	0.002	0.001	-0.000	0.005	0.004
	[0.417]	[0.416]	(0.005)	(0.006)	(0.004)	(0.005)	(0.005)
Fourth-year	0.147	0.147	0.001	-0.005	0.005	-0.001	-0.007
	[0.354]	[0.354]	(0.005)	(0.006)	(0.006)	(0.005)	(0.005)
Fifth-year	0.181	0.181	-0.004	-0.004	-0.001	0.004	-0.002
	[0.385]	[0.385]	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)
Rural 5-year program	0.068	0.069	0.000	-0.011	-0.005	0.001	0.003
	[0.252]	[0.253]	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)

Variable	All	C	T1-C	T2-C	T3-C	T4-C	T5-C
5-year program	0.860	0.859	0.005	0.005	0.003	0.001	0.002
	[0.347]	[0.348]	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
8-year program	0.072	0.073	-0.010	0.002	-0.002	-0.003	-0.006
	[0.258]	[0.259]	(0.008)	(0.007)	(0.007)	(0.007)	(0.006)
Father's ISEI level	0.639	0.638	0.002	-0.003	0.003	0.005	0.002
	[0.480]	[0.481]	(0.003)	(0.004)	(0.003)	(0.004)	(0.004)
Mother's ISEI level	0.658	0.658	-0.001	-0.006	-0.002	0.005	0.002
	[0.474]	[0.474]	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)
Father's education	0.274	0.276	-0.001	0.004	-0.004	-0.006	-0.004
	[0.446]	[0.447]	(0.003)	(0.005)	(0.004)	(0.004)	(0.004)
Mother's education	0.223	0.223	0.001	0.003	-0.003	-0.003	-0.003
	[0.416]	[0.416]	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Low family income	0.644	0.644	0.004	-0.006*	-0.005	0.001	0.004
	[0.479]	[0.479]	(0.003)	(0.003)	(0.004)	(0.003)	(0.004)
NCEE score	549.987	550.189	-0.000**	0.000	0.000	-0.000	-0.000
	[61.272]	[61.311]	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
A family of doctors	0.318	0.315	0.005	0.005	0.006	0.003	0.001
	[0.466]	[0.465]	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Higher admission	0.363	0.363	0.001	-0.003	-0.001	0.002	-0.001
expectation	[0.481]	[0.481]	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Senior high school	0.481	0.483	-0.003	-0.002	-0.002	-0.004	-0.002
type	[0.500]	[0.500]	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Ideal career	0.534	0.534	0.001	0.001	-0.003	-0.001	-0.001
(1=doctor)	[0.499]	[0.499]	(0.004)	(0.003)	(0.003)	(0.004)	(0.003)
Higher parents'	0.147	0.148	-0.002	-0.009	0.000	0.003	-0.003
expectation	[0.354]	[0.355]	(0.006)	(0.005)	(0.006)	(0.005)	(0.005)
Schools' location	0.384	0.386	-0.007*	-0.001	-0.000	-0.004	-0.003
(eastern)	[0.486]	[0.487]	(0.003)	(0.004)	(0.004)	(0.003)	(0.003)

Variable	All	C	T1-C	T2-C	T3-C	T4-C	T5-C
Schools' location	0.280	0.277	0.011***	0.001	0.005	0.007**	-0.001
(central)	[0.449]	[0.447]	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Schools' location	0.337	0.337	-0.003	0.000	-0.004	-0.002	0.004
(western)	[0.473]	[0.473]	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
University attribution	0.697	0.702	-0.005	-0.006	-0.006	-0.009***	-0.006
	[0.460]	[0.458]	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)
University level	0.141	0.141	-0.002	0.006	0.008	-0.005	-0.002
	[0.348]	[0.348]	(0.005)	(0.005)	(0.005)	(0.004)	(0.006)
Career training	0.293	0.295	-0.007**	-0.002	-0.003	-0.003	-0.005
	[0.455]	[0.456]	(0.004)	(0.004)	(0.005)	(0.004)	(0.004)
Observations	213326	35753	35777	36076	35301	34461	35958

Table A2 Robustness check: recode dependent variable

	Endline Career Commitment (Neutral = 1)			
	(1)	(2)	(3)	(4)
T1: Significance of the profession	0.003**	0.003***	0.003***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
T2: Social respect for doctors	0.004***	0.004***	0.004***	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)
T3: Altruism identity and role model	0.002	0.002*	0.002*	0.002*
	(0.001)	(0.001)	(0.001)	(0.001)
T4: Salary level	0.004**	0.004***	0.006***	0.004***
	(0.001)	(0.001)	(0.002)	(0.001)
T5: Peer retention rate	0.003***	0.003***	0.003***	0.002**
	(0.001)	(0.001)	(0.001)	(0.001)
Low family income			0.003***	
			(0.001)	
Low family income			-0.004*	
× Salary level (T4)			(0.002)	
Overestimate dropout				-0.015***
				(0.001)
Overestimate dropout				0.004
× Peer retention rate (T5)				(0.003)
Baseline Career Commitment		0.102***	0.102***	0.100***
		(0.002)	(0.002)	(0.002)
Control group mean	0.972	0.972	0.972	0.972
Other Controls	NO	YES	YES	YES
School FE	NO	YES	YES	YES
R^2	0.000	0.066	0.066	0.068
Observations	213,326	213,326	213,326	213,326

Notes: Control variables are the same as that in Table 2. Standard errors are clustered at the school level shown in the parentheses. Significant at *10%, **5%, and ***1%..

B. Construction of four indices for baseline characteristics

Intrinsic motivation index:

The intrinsic motivation is constructed from the responses to the following five survey questions.

1.Achieved good grades in subjects related to this major during high school.

2.Interested in medical-related majors.

3.Confident in achieving success in this field.

4.Believe that being a doctor is respected.

5.Medicine is a sacred and noble profession dedicated to saving lives.

Survey respondents evaluate each statement a 5-Likert scale from 1 to 5: 1=strongly disagree, 2=disagree, 3=neutral, 4=agree, 5=strongly agree. The intrinsic motivation index is the sum of all five evaluations.

Extrinsic motivation index:

Similarly, extrinsic motivation index is the sum of Likert scale (1-5) evaluations of the following three statements.

1.Studying medicine offers good employment opportunities.

2.My parents, relatives, or teachers encouraged (or required) me to choose a medical major.

3.Studying medicine can provide convenience for me, my family, and friends to access medical resources in the future.

Major satisfaction index:

The major satisfaction index is the sum of the likert scale (1-5 or 1-3) evaluations of the following four questions:

1.Please evaluate your satisfaction with your major admission. (1=least satisfy, 5=most satisfy)

2.Before studying your current major, how well did you understand this major? (1= do not understand at all, 5= understand very well)

3.When choosing a university major, what did you think about the working environment of medical professionals in our country? (1=very bad, 5= very good)

4.Based on your actual situation, estimate the gap between your university admission results and your expected goals. (1=worse than expectation, 2=as expected, 3=better than expectation).

College experience index:

The college experience index summarize the following responses in Likert scale 1-5:

1.Are you satisfied with the course design of your major, including Natural science courses, Humanities and social sciences courses, Scientific methodology courses, Biomedical courses, Public health courses, Traditional Chinese medicine courses, Clinical medicine courses. (7 questions in total.)

2.Are you satisfied with the other aspects of the institution, including Facilities, Electronic resources supporting student self-study, Elective course system, Grading system, Award and recognition system, Student representation in relevant school committees, Academic guidance and counseling, Psychological counseling services, Career planning and guidance, Clubs and extracurricular activities. (10 question in total.)

C. Original Treatment Materials

Treatment 1: Significance of the profession

医务人员是国家最宝贵的资源

詹启敏

（中国工程院院士、北京大学常务副校长、医学部主任、全国医学教育发展中心主任）



一场突如其来的疫情突袭了全球，给各国人民健康和经济发展带来了巨大的创伤，也显示了医学和医学教育在国家安全体系中的重要地位，医学和医学教育是提供全民健康和保证。

医务人员是国家最宝贵的资源。中国自古以来便有"大医精诚"的理念。我们需要进一步凝聚社会共识：医务人员是国家最宝贵的资源，是守卫人民生命安全和健康的卫士，应提倡全社会尊医重卫、崇尚医疗健康行业，形成共建健康中国的良好氛围。

Original version

Healthcare professionals are the nation's most valuable resource

Qimin Zhan

(Academician of the Chinese Academy of Engineering, Executive Vice President of Peking University, Dean of the Faculty of Medicine, Director of the National Centre for Health Professions Education Development)



An unexpected pandemic struck the world, causing significant harm to the health and economic development of people across various countries. It also underscored the importance of medicine and medical education in the national security system, as they are essential for ensuring the health of the population.

Healthcare professionals are the nation's most valuable resource. Since ancient times, China has upheld the concept of "Great Physicians Possess Sincerity and Integrity (Da Yi Jing Cheng)." We need to further solidify social consensus: healthcare professionals are the country's most valuable resource and the guardians of people's life and health. It is crucial to advocate for respect towards the medical profession across society, to esteem the healthcare industry, and to foster a favorable environment for building a healthy China together.

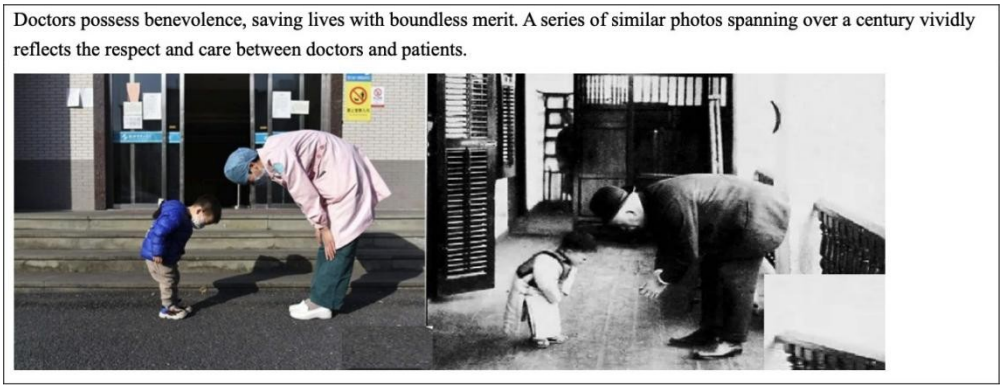
Translated version

We provided ChatGPT with this intervention message to predict its effectiveness in reducing medical students' attrition rates. The prompt was "Supposing you are an undergraduate medical student in China, you have a probability of 13% of thinking of dropping out of the medical career. How would your attrition probability change if you saw 5 different pieces of information below? Give the numbers for your proportional change for each." Starting with a baseline dropout probability of 13%, ChatGPT estimated a 3% reduction in dropout probability (to 10%) if students were exposed to a message emphasizing the importance of the medical profession.

Treatment 2: Social respect for doctors



Original version



Translated version

For the second intervention, ChatGPT was asked to predict the impact of a message highlighting the respect and care between doctors and patients (however, this diminishes the potential impact of the visual presentations). Starting from a baseline dropout probability of 13%, ChatGPT estimated a 2% reduction (to 11%).

Treatment 3: Altruism identity and role model

自从 COVID-19 以来，医生与护士和其他在前线工作的医护人员一起，成为了"最美逆行者"、名副其实的英雄——他们冒着自己的生命危险为他人服务！谁言医者皆文弱，不破疫情誓不回！全国各地千千万万的医生，在本次疫情中体现出作为医者的担当，就像钟南山院士所说，**他相信他们都是英雄！**



《人民日报》曾这样评价钟南山：既有国士的担当，又有战士的勇猛。他回应得最多的一句话是"我不过是一个看病的大夫"。穿上这身白衣，便选择全力以赴。**以钟南山为榜样，全国已有上万名年轻医务工作者奔赴湖北。**他们恪尽职守、一丝不苟，只因医者仁心，不辱使命。

Original version

Since COVID-19, doctors, along with nurses and other frontline healthcare workers, have become the "most beautiful counter-marchers" and true heroes—they risk their own lives to serve others! Who says that doctors are all frail scholars? They vow not to return until the pandemic is defeated! Thousands of doctors across the country have demonstrated the responsibility of being a physician during this pandemic. As Academician Zhong Nanshan said, he believes they are all heroes!

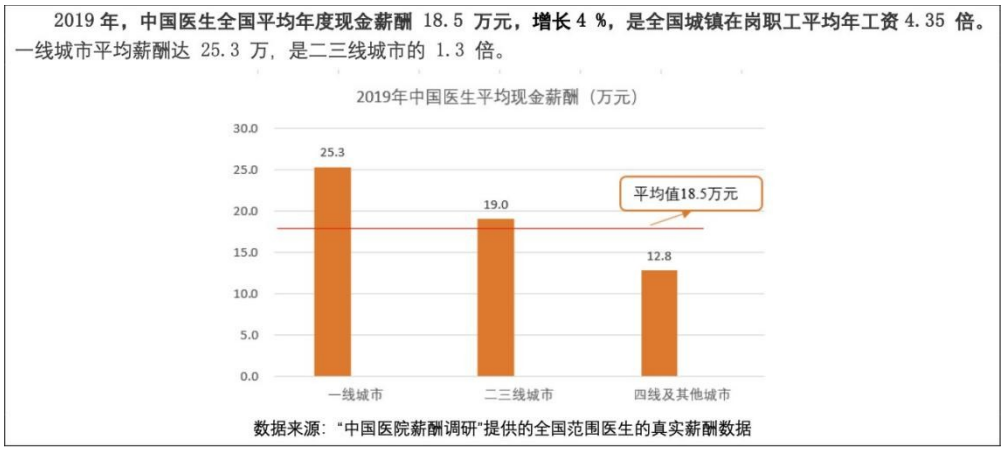


The People's Daily once praised Zhong Nanshan as having both the responsibility of a national figure and the courage of a warrior. The phrase he has most often responded with is, "I am merely a doctor treating patients." Wearing this white coat is a choice to give one's all. Inspired by Zhong Nanshan, over 10,000 young medical workers nationwide have already gone to Hubei. They are dedicated and meticulous in their duties, simply because of their benevolent hearts and their commitment to not failing their mission as healers.

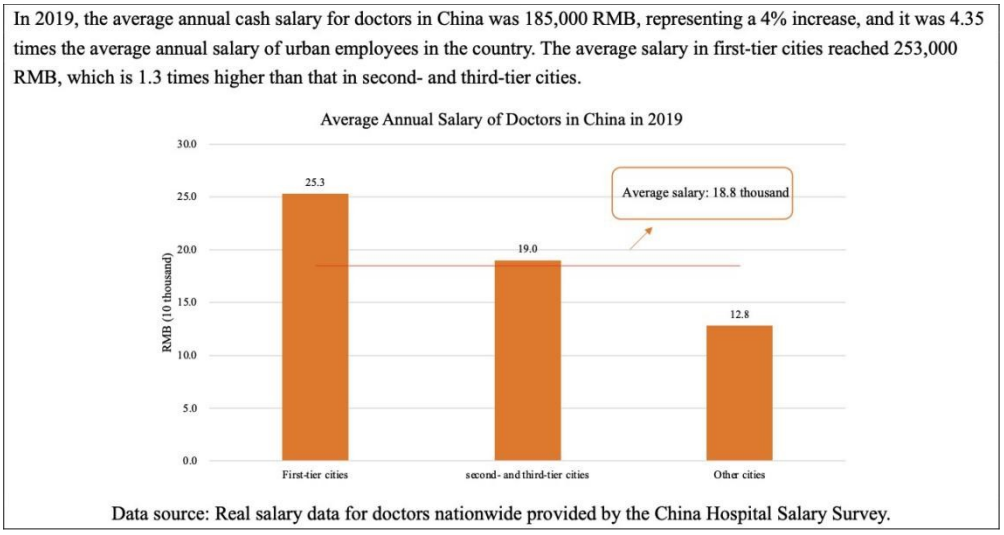
Translated version

For the third intervention, which focused on the heroic role of doctors, ChatGPT predicted a stronger impact due to the emotional appeal. From the 13% baseline, it estimated a 4% reduction in dropout probability (to 9%).

Treatment 4: Salary Level



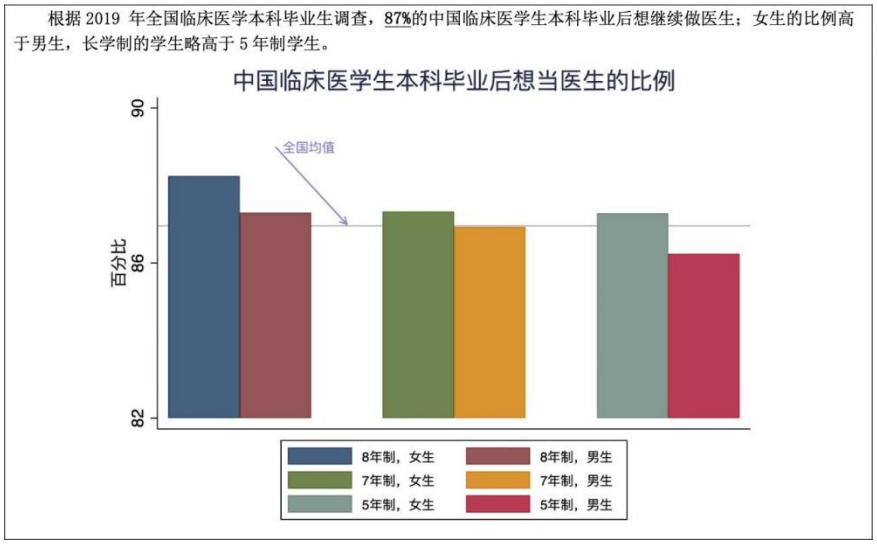
Original version



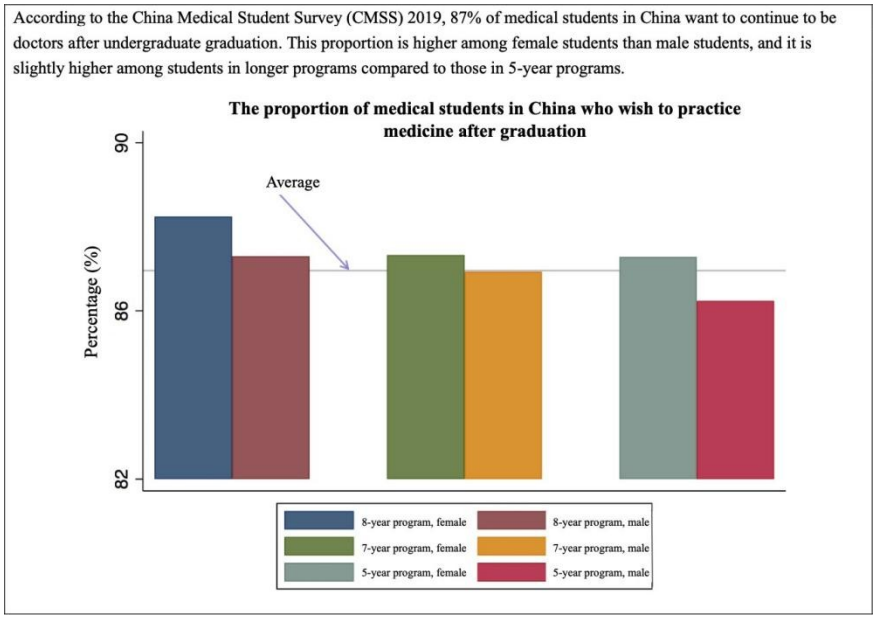
Translated version

For the fourth intervention, centered on financial incentives, ChatGPT estimated a smaller impact with a 1% reduction in dropout probability (to 12%) starting from the 13% baseline.

Treatment 5: Peer retention rate



Original version



Translated version

For the fifth intervention, highlighting peer commitment to the medical profession, ChatGPT estimated a 2% reduction in dropout probability (to 11%) from the baseline of 13%.